Semantic Response Ranking for Open-Domain Chatbots

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Embedding-based Candidate Selection with MRR & Recall@k Evaluation

Motivation

Modern chatbots generate many candidate replies but need to select the most coherent one automatically.
Goal: given a conversation history (context), rank candidate responses by semantic similarity and measure ranking quality.
Pipeline:
- Embed context & candidates with a sentence transformer
- Rank by cosine similarity: sim(ctx, cand) = v_ctx · v_cand (L2-normalised)
- Evaluate: MRR, Recall@k vs. random baseline

Datasets

Dataset	Description	Total (test)	Tested
Synthetic	Hand-crafted (10 domains)	50	50
SODA	It is a large dataset (1M+ dialogues) for everyday social conversations (open-domain chat). Purpose: to train a model to carry out natural conversations. It helps improve coherence, "human-likeness," and contextual awareness in dialogue.	148,968	300
BlendedSkillTalk	A dataset for training a model to blend different conversational skills within a single dialogue. Skills include: empathy, knowledge (factual information), personality, and engaging conversation.	980	300
HH-RLHF	A dataset for training via preferences (RLHF). It is designed to identify the most helpful and safe responses.	8,552	300

Experimental Results for pool size 10

Dataset	MRR	R@1	R@2	R@3	R@5
Synthetic	**0.990**	**0.980**	**1.000**	**1.000**	**1.000**
HH-RLHF	**0.911**	0.873	**0.907**	**0.933**	**0.970**
BlendedSkillTalk	0.834	0.737	0.863	**0.917**	**0.973**
SODA	0.760	0.647	0.763	0.850	**0.913**
Random baseline	*0.293*	*0.100*	*0.200*	*0.300*	*0.500*

Comparison pool size 10 vs. pool size 50

Dataset	MRR (pool=10)	MRR (pool=50)	R@1 (pool=10)	R@1 (pool=50)
Synthetic	**0.990**	**0.990**	**0.980**	**0.980**
HH-RLHF	**0.911**	0.863	**0.873**	**0.833**
BlendedSkillTalk	0.834	0.671	0.737	**0.550**
SODA	0.760	0.579	0.647	0.470
Random baseline	*0.293*	*0.090*	*0.100*	*0.020*

HH-RLHF dropped slightly (0.91 → 0.86) — structured dialogues are still well distinguished.
SODA dropped significantly (0.76 → 0.58) — with a larger candidate pool, it becomes much harder to identify the correct "empty" response.
The random baseline fell from 0.293 to 0.090 — the task became ~3× harder, so the relative lift increased even though absolute metrics declined.

Failure Analysis: Semantics vs. Pragmatics

Core limitation: cosine similarity measures semantic closeness, not conversational appropriateness.
Two texts can be semantically similar yet one may be a poor response in context — embeddings cannot detect this.
Why? The reply is a social agreement phrase — it contains no topic words (birds, nature, trees). Embeddings capture topic, not pragmatic fit.

SODA Failure Case (gold rank: 10/10, score: −0.005)

- [1] Hey, thanks for coming out to help me…
- [2] No problem. I love being outdoors…
- [3–9] …planting trees, protecting nature, birdhouses…
- [13] I've put up birdhouses to attract wildlife.
- [14] That's great! I'm sure the birds appreciate it.
- **[15] GOLD: "Yeah, I hope so!" , ranked LAST (10/10)**

Future Work: Cross-Encoder Reranking

Current approach (bi-encoder) encodes context and candidate independently - it cannot model their interaction.
Cross-encoders process the pair (context, candidate) jointly through a transformer, capturing fine-grained relevance signals that bi-encoders miss.
Expected metric improvements: Hardest cases (pragmatic replies): significant gain expected
However, cross-encoders are 10–100× slower at inference time.

Conclusions

- Performance degrades on open-domain chitchat (SODA) due to pragmatic vs. semantic gap.
- Structured assistant dialogues (HH-RLHF) best suited for semantic ranking (R@1 = 87.3%).
- Synthetic data reveals the upper bound of the method; real data reveals its fundamental limitations.
- Cross-encoder reranking identified as the primary next step to address pragmatic failures.